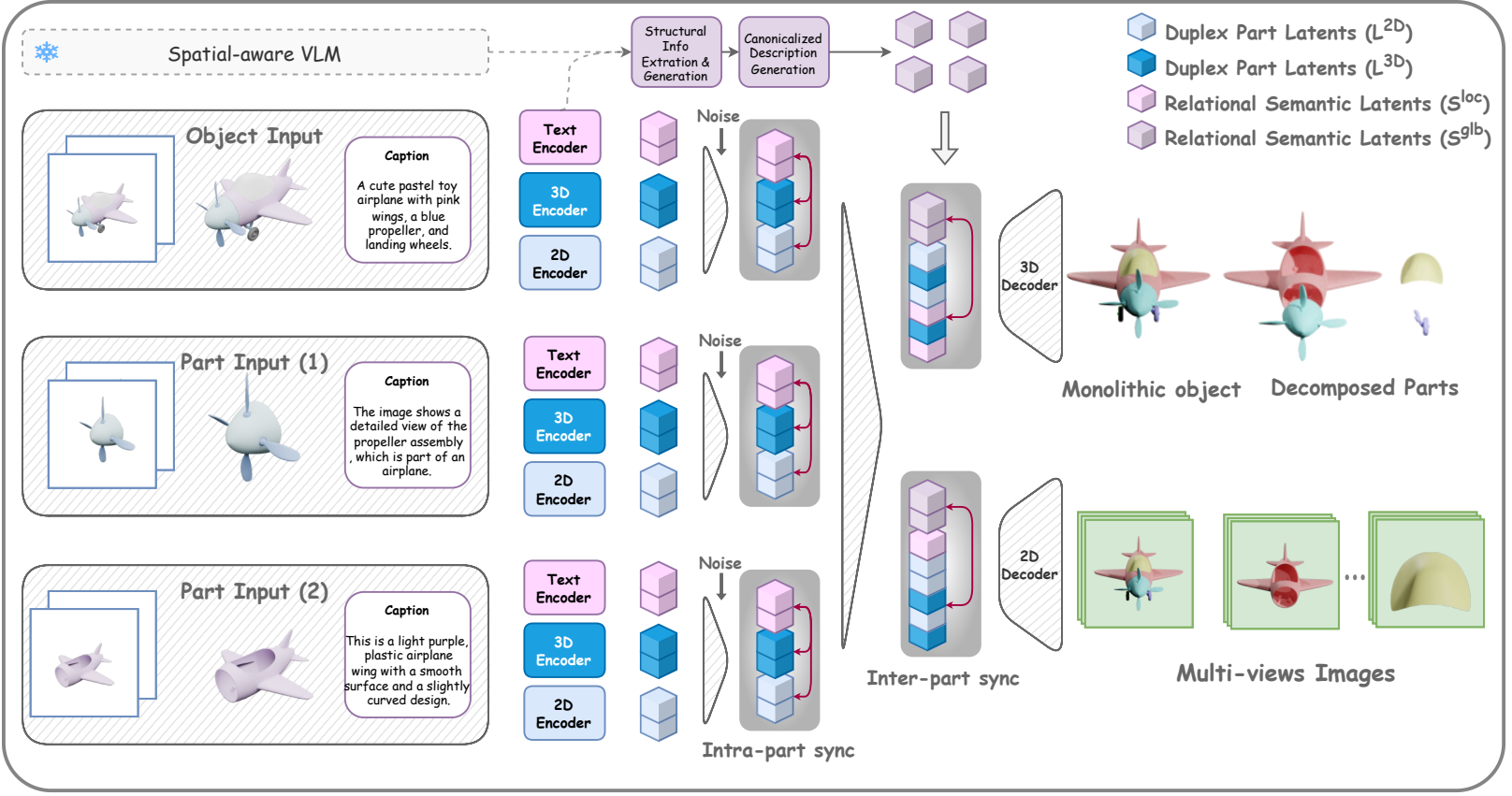
Spatial-aware VLM
Structural Info Extration & Generation
Canonicalized Description Generation
Duplex Part Latents (L^{2D})
Duplex Part Latents (L^{3D})
Relational Semantic Latents (S^{loc})
Relational Semantic Latents (S^{glb})
Object Input
Caption
A cute pastel toy airplane with pink wings, a blue propeller, and landing wheels.
Part Input (1)
Caption
The image shows a detailed view of the propeller assembly , which is part of an airplane.
Part Input (2)
Caption
This is a light purple, plastic airplane wing with a smooth surface and a slightly curved design.
Text Encoder
3D Encoder
2D Encoder
Noise
Intra-part sync
Inter-part sync
3D Decoder
2D Decoder
Monolithic object
Decomposed Parts
Multi-views Images